Planning & Management

1.1 Project Overview

Project Name

HR Analytics & Employee Attrition Analysis Using Power BI

Background & Problem Statement

Employee retention is a major challenge for organizations, as losing employees can lead to increased recruitment costs, loss of experience, reduced productivity, and additional pressure on existing employees.

Organizations need to understand not only how many employees leave but also the factors and patterns associated with their decisions to leave.

This project is based on an HR dataset containing information about 1,470 employees. Initial exploration shows that approximately 237 employees have left the company, representing an attrition rate of 16.1%.

The main problem is that the organization needs a clearer understanding of employee attrition patterns and the workplace factors associated with them. These may include compensation, overtime, job satisfaction, work environment, performance, training opportunities, promotions, and career development.

Therefore, this project aims to investigate these factors, identify employee groups with higher attrition rates, and provide insights that can support better HR decisions.

Project Objectives

The main objectives of this project are:

1. **Analyze Workforce Structure:** Understand the company's workforce distribution across departments, job roles, age groups, gender, and employee tenure.
2. **Measure Employee Attrition:** Calculate the overall attrition rate and identify the departments, job roles, and employee groups with the highest attrition rates.
3. **Investigate Attrition Factors:** Explore the relationships between employee attrition and salary, overtime, business travel, job satisfaction, work-life balance, and other workplace conditions.
4. **Evaluate Employee Performance:** Analyze performance ratings, training activities, salary increases, promotions, and differences between employee self-ratings and manager ratings.

5. **Identify Retention Challenges:** Discover employee groups that may be more vulnerable to leaving the organization, particularly those experiencing low satisfaction or limited career growth.
6. **Develop an Interactive Dashboard:** Build a comprehensive four-page Power BI dashboard that presents key HR metrics and allows users to explore the data through interactive visualizations and filters.
7. **Provide Data-Driven Recommendations:** Translate analytical findings into practical recommendations that can help improve employee satisfaction and retention.

Expected Business Value

This project is expected to provide several business benefits:

- **Improved Employee Retention:** Help HR identify employee groups with higher attrition rates and develop targeted retention strategies.
- **Better HR Decision-Making:** Provide management with reliable metrics and insights to support strategic workforce planning.
- **Reduced Recruitment Costs:** Support efforts to reduce unnecessary employee turnover and the costs associated with hiring and onboarding replacements.
- **Enhanced Employee Satisfaction:** Highlight potential issues related to job satisfaction, work environment, overtime, and work-life balance.
- **Performance & Career Development:** Help identify opportunities to improve employee development through training, promotions, and performance evaluation.
- **Data-Driven HR Strategy:** Transform raw employee data into actionable insights through an interactive dashboard, enabling HR teams to monitor workforce trends and make more informed decisions.

Ultimately, the project aims to help the organization move from reactive employee management toward a more proactive, data-driven approach to workforce planning and employee retention.

1.2 Project Scope

1.2.1 In Scope

The project focuses on analyzing employee data to understand workforce composition, investigate attrition patterns, and identify factors associated with employee turnover.

The analysis covers the following areas:

1. Workforce & Attrition Analysis

- Analyze the total workforce and employee distribution across departments and job roles.
- Examine employee demographics, including age and gender.
- Calculate total attrition and overall attrition rate.

- Identify departments, job roles, age groups, and tenure groups with higher attrition rates.

2. Employee Satisfaction & Work Environment

- Analyze job satisfaction, environment satisfaction, and relationship satisfaction.
- Examine work-life balance across different employee groups.
- Compare satisfaction levels between employees who stayed and those who left.
- Identify groups experiencing low satisfaction and higher attrition rates.

3. Compensation & Working Conditions

- Analyze salary distribution across departments and job roles.
- Investigate the relationship between salary levels and employee attrition.
- Examine overtime and business travel patterns.
- Analyze distance from home and other available working-condition indicators.
- Explore whether compensation and working conditions are associated with employee retention.

4. Performance, Training & Career Growth

- Analyze employee performance ratings across departments and job roles.
- Compare employee self-ratings with manager ratings.
- Examine training frequency and its relationship with employee performance.
- Investigate salary increases, promotion history, and career development.
- Identify employees with missing performance evaluation records.

5. Data Quality Validation

- Identify missing values, duplicate records, and inconsistencies.
- Validate Employee IDs and relationships between tables.
- Investigate differences in employee counts across datasets.
- Review inconsistencies between training opportunities and completed training sessions.
- Verify the level of detail in performance and rating records before combining them with employee data.

1.2.2 Out of Scope

To maintain a clear project focus, the following activities are not included:

- Predicting the exact probability that a specific employee will leave using machine learning models.
- Building an automated employee monitoring or early-warning system.
- Conducting employee interviews or collecting new survey responses.
- Evaluating individual employees' personal behavior beyond the information available in the dataset.
- Implementing changes to company HR policies or recruitment processes.

- Performing financial forecasting or calculating the exact financial cost of every employee departure.
- Establishing direct causal relationships between workplace factors and employee attrition.

1.2.3 Project Limitations

The analysis depends on the quality, completeness, and structure of the available datasets. The following limitations should be considered:

- **Incomplete Performance Records:** The performance evaluation dataset contains approximately 1,280 records compared with 1,470 employees, requiring further investigation into missing evaluations.
- **Training Data Inconsistencies:** Some records show differences between training opportunities and completed training sessions, and the meaning of these fields must be validated.
- **Multiple Rating Records:** Self-rating and manager-rating records may contain multiple entries per employee, so their level of detail must be verified before analysis.
- **Limited Historical Context:** The available data may not provide enough historical information to track changes in employee satisfaction, performance, or compensation over time.
- **Correlation vs. Causation:** The analysis can identify relationships between employee attrition and workplace factors, but it cannot independently prove that these factors cause employees to leave.
- **Data Availability:** Some business questions may be limited by missing, incomplete, or unavailable information.

1.3 Business Requirements & Questions

1.3.1 Business Requirements

The organization requires an interactive HR analytics dashboard that provides a clear understanding of workforce structure, employee attrition, satisfaction, compensation, performance, and career development.

The dashboard should:

1. Present key HR performance indicators through clear and measurable KPIs.
2. Allow users to analyze employee attrition across different workforce segments.
3. Highlight employee groups with higher attrition rates.
4. Explore relationships between attrition and workplace conditions.
5. Provide insights into employee performance, training, and career growth.

6. Enable interactive filtering and comparisons across departments, job roles, age groups, and other relevant dimensions.
7. Support data-driven HR decisions through meaningful visualizations and actionable insights.

1.3.2 Business Questions

The following business questions will guide the data analysis and dashboard development.

A. Workforce Overview

Objective: Understand the overall workforce structure and identify the employee groups most affected by attrition.

1. What is the total workforce size, and how many employees have left the company?
2. What is the overall employee attrition rate?
3. Which departments and job roles have the highest and lowest attrition rates?
4. Which age groups and gender groups experience the highest attrition rates?
5. Are employees with shorter tenure more likely to leave the company?

B. Attrition Drivers

Objective: Investigate the workplace factors and employee conditions associated with higher attrition rates.

1. Is overtime associated with higher employee attrition?
2. Do employees in lower salary bands have higher attrition rates?
3. Is business travel frequency associated with employee attrition?
4. Are employees who live farther from work more likely to leave?
5. Are employees with lower job satisfaction or poor work-life balance more likely to leave?
6. Are employees with lower performance ratings more likely to leave?
7. Is a longer period without promotion associated with higher attrition?

C. Performance & Career Growth

Objective: Evaluate employee performance and understand how training, compensation growth, and career development relate to employee retention.

1. How does employee performance vary across departments and job roles?
2. Is higher performance associated with higher salary increases?
3. Does training frequency vary by employee performance, and is more training associated with better performance?
4. How large is the gap between employee self-ratings and manager ratings, and which groups have the largest gaps?

5. Are employees without recorded performance evaluations more likely to leave?
6. How does promotion history vary across employee groups with different attrition statuses?

D. Data Quality & Retention Insights

Objective: Validate the reliability of the available data and identify employee groups that require greater attention from HR.

1. Are there inconsistencies between training opportunities and the number of training sessions completed?
2. Which employee groups have both low satisfaction and high attrition rates?
3. Which combinations of job role, salary, tenure, and satisfaction are associated with the highest attrition rates?
4. Are missing performance evaluations concentrated in specific departments or employee groups?
5. Which employee groups and workplace factors should HR prioritize to improve employee retention?

1.3.3 Expected Dashboard Outcomes

The answers to these questions will be presented through four interactive Power BI dashboard pages:

1. **HR Overview & Attrition:** Workforce distribution, employee headcount, and attrition patterns.
2. **Satisfaction & Work Environment:** Employee satisfaction levels and their relationship with attrition.
3. **Compensation & Work Conditions:** Salary, overtime, business travel, and other working conditions.
4. **Performance, Training & Career Growth:** Performance evaluations, training activities, salary increases, and promotions.

The final dashboard will provide management with a structured view of workforce trends and help identify areas where HR interventions may improve employee retention.

1.4 Team Structure & Responsibilities

1.4.1 Team Structure

The project is carried out by a team of six members, working collaboratively to complete the HR data analysis, develop the Power BI dashboard, and deliver meaningful business insights.

Team Leader:

- Farah Abd Elmonem

Team Members:

- Farah Wael
- Ganna Ashraf
- Salma Gabr
- Samar Ahmed
- Habiba Abdelbaie

1.4.2 Roles & Responsibilities

The project responsibilities are distributed among team members according to the main project phases to ensure efficient task execution and clear ownership.

Team Member	Assigned Responsibilities
Farah Abd Elmonem (Team Leader)	Project coordination, task distribution, progress monitoring, dashboard development, data analysis, and final review of project deliverables.
Farah Wael	Data cleaning, data preparation, handling missing values and inconsistencies, and preparing datasets for analysis.
Ganna Ashraf	Data cleaning, data preparation, data validation, and supporting the preparation of datasets for analysis.
Habiba Abdelbaie	Data cleaning, data preparation, identifying data quality issues, and ensuring datasets are ready for analysis.
Samar Ahmed	Power BI dashboard development, data visualization, analysis, and presenting key findings.
Salma Gabr	Power BI dashboard development, data visualization, analysis, and supporting the preparation of insights and recommendations.

1.4.3 Task Ownership

The work is divided into two main teams:

1. Data Cleaning & Preparation Team

- Farah Wael
- Ganna Ashraf
- Habiba Abdelbaie

Main responsibilities:

- Explore and understand the available HR datasets.
- Identify and handle missing values and duplicate records.
- Investigate inconsistencies across tables.

- Validate Employee IDs and table relationships.
- Prepare clean and structured datasets for analysis.

2. Data Analysis & Dashboard Development Team

- Farah Abd Elmonem
- Samar Ahmed
- Salma Gabr

Main responsibilities:

- Analyze workforce and employee attrition patterns.
- Develop DAX measures and calculate key HR metrics.
- Design and develop interactive Power BI dashboard pages.
- Create visualizations that address the business questions.
- Identify meaningful insights and prepare business recommendations.
- Review and finalize the dashboard and project deliverables.

1.4.4 Collaboration & Communication

The team follows a collaborative working approach to ensure smooth coordination and consistent progress throughout the project.

The collaboration process includes:

- **Task Distribution:** The team leader assigns responsibilities based on project requirements and each member's assigned role.
- **Regular Communication:** Team members communicate through group discussions to share updates, ask questions, and resolve challenges.
- **Progress Monitoring:** Tasks are reviewed regularly to track completion and identify delays or difficulties.
- **File Sharing & Version Control:** Project files, datasets, documentation, and deliverables are shared through common storage and the GitHub repository.
- **Team Review:** Team members collaborate to review data quality, validate analytical results, and ensure consistency across dashboard pages.
- **Final Integration:** All completed tasks are combined and reviewed by the team leader to ensure the final deliverables meet the project objectives.

Team Management Approach:

The team leader, Farah Abd Elmonem, oversees project coordination, monitors progress, ensures communication between the two working groups, and reviews the final project outputs. All team members contribute to achieving the shared project objectives.

1.5 Project Methodology & Workflow

1.5.1 Project Methodology

The project follows a structured data analytics methodology to transform raw HR data into meaningful insights that support employee retention and workforce management decisions.

The methodology consists of the following phases:

1. Data Understanding & Exploration

- Identify the available datasets and understand their structure.
- Review tables, columns, data types, and key attributes.
- Understand the meaning of HR metrics and business-related fields.
- Identify the main business questions that the analysis needs to answer.

2. Data Cleaning & Preparation

- Identify and handle missing values and duplicate records.
- Correct data inconsistencies and formatting issues.
- Validate Employee IDs and investigate differences in employee counts across tables.
- Review the training and performance evaluation records.
- Prepare clean datasets for further analysis.

3. Data Modeling

- Identify fact and dimension tables where applicable.
- Define relationships between tables using Employee ID and other relevant keys.
- Validate relationship cardinality and filter directions.
- Ensure the data model supports accurate calculations and analysis.

4. Data Analysis & DAX Development

- Create calculated measures and key HR performance indicators using DAX.
- Calculate total employees, total attrition, attrition rate, and average age.
- Analyze attrition across departments, job roles, demographics, and tenure.
- Investigate relationships between attrition and satisfaction, salary, overtime, performance, and career growth.
- Validate analytical results before using them in visualizations.

5. Dashboard Design & Development

- Develop an interactive four-page Power BI dashboard.
- Select appropriate visualizations based on each business question.
- Design KPI cards, charts, tables, and interactive slicers.

- Maintain consistency in layout, colors, formatting, and user experience.
- Test interactions and ensure that filters work correctly across visuals.

6. Insights & Recommendations

- Interpret the results and identify important workforce patterns.
- Highlight employee groups with higher attrition rates.
- Identify potential retention challenges.
- Develop practical, data-driven recommendations for HR management.

7. Testing & Final Documentation

- Review KPI calculations and dashboard accuracy.
- Validate data relationships and visual interactions.
- Check that the dashboard answers the defined business questions.
- Document the methodology, findings, and recommendations.
- Prepare the final project deliverables.

1.5.2 Project Workflow

The project follows a sequential workflow in which each phase builds on the outputs of the previous phase.

Workflow:

1. Data Collection
2. Data Understanding & Exploration
3. Data Cleaning & Preparation
4. Data Modeling
5. DAX Measures & Calculations
6. Exploratory Data Analysis
7. Dashboard Design & Development
8. Testing & Validation
9. Insights & Business Recommendations
10. Final Documentation & Presentation

Tools & Technologies:

- Microsoft Excel: Initial data exploration and dataset handling.
- Power Query: Data cleaning and transformation.
- Power BI: Data modeling, analysis, and interactive dashboard development.
- DAX: Creating measures and calculating HR metrics.
- GitHub: Project file management, documentation, and collaboration.

1.6 Deliverables & Success Criteria

1.6.1 Project Deliverables

The project is expected to produce the following deliverables:

1. Cleaned & Prepared Datasets

- Clean and structured HR datasets.
- Identified and documented data quality issues.
- Validated employee identifiers and table relationships.

2. Data Model

- Organized data model with appropriate fact and dimension tables where applicable.
- Validated relationships between datasets.
- A structure that supports accurate HR calculations.

3. HR Analytics Power BI Dashboard

An interactive dashboard consisting of four main pages:

- **Page 1 – HR Overview & Attrition:** Workforce KPIs, employee distribution, and attrition analysis.
- **Page 2 – Satisfaction & Work Environment:** Satisfaction metrics and their relationship with employee attrition.
- **Page 3 – Compensation & Work Conditions:** Salary analysis, overtime, business travel, and other working conditions.
- **Page 4 – Performance, Training & Career Growth:** Performance evaluations, training activities, salary increases, and promotions.

4. Business Insights & Recommendations

- Key findings from the employee attrition analysis.
- Identification of workforce segments with higher attrition rates.
- Potential retention challenges.
- Practical recommendations to support HR decision-making.

5. Project Documentation

- Project planning and management.
- Data understanding and preparation.
- Data modeling and analysis methodology.
- Dashboard explanation and key findings.
- Final conclusions and recommendations.

6. GitHub Repository

- Project files and supporting resources.
- Documentation and relevant project materials.
- Power BI project file, where applicable.

1.6.2 Success Criteria

The success of the project will be evaluated based on the following criteria:

Success Area	Evaluation Criteria
Data Quality	Missing values, duplicates, and inconsistencies are identified and handled appropriately.
Data Accuracy	Employee counts, attrition rates, and other KPIs are correctly calculated and validated.
Data Modeling	Relationships between tables are properly established and tested.
Business Requirements	The dashboard addresses the defined business questions.
Dashboard Functionality	Visuals, slicers, filters, and interactions work correctly.
Dashboard Design	The dashboard has a clear, consistent, and user-friendly layout.
Analytical Value	The analysis identifies meaningful patterns and relationships within the HR data.
Business Recommendations	Findings are translated into practical recommendations that support employee retention.
Project Completion	All agreed deliverables are completed and documented.

1.6.3 Expected Final Outcome

The final outcome will be an interactive Power BI HR Analytics Dashboard that transforms employee data into meaningful business insights.

The dashboard will help HR management understand workforce composition, monitor employee attrition, investigate potential retention challenges, and evaluate the relationships between employee satisfaction, compensation, performance, and career development.

Ultimately, the project aims to support more informed HR decisions and contribute to developing effective employee retention strategies.